

# Lab Workbook

---

*Fourteen guided studios, response sheets and device-free alternatives*

Prepared for Yasin Kaymaz, PhD | Version 1.0 | 14 September 2026

This is a teaching package, not a clinical, legal or institutional accreditation document. Scientific sources are identified by [Sxx] in the Course Source Register.

## How to use this workbook

Each studio is designed for 45 minutes. Use the matching notebook for calculations and this workbook for planning, discussion or a paper submission. Do not write a long report for every studio. Only the eight A1 weeks are completion-graded; W05 and W09 launch D1 and D2; other activities are formative unless the published course policy states otherwise.

The same grading opportunity must be available for device-free work. A documented execution error plus a valid paper calculation and interpretation can demonstrate learning. Ask for help early. All simulation data are synthetic and cannot be presented as research findings.

| Minutes | Activity                                |
|---------|-----------------------------------------|
| 0-5     | Read the question; write a prediction.  |
| 5-10    | Set up or inspect the printed records.  |
| 10-30   | Perform the guided investigation.       |
| 30-40   | Answer Q1-Q3 and compare evidence.      |
| 40-45   | Complete the check/disclosure and save. |

## W01 | From a biological question to defensible evidence

Calculate and explain a sequence summary while preserving provenance.

### Before you begin

Recall: genes can produce RNA or protein products; RNA counts are not direct protein measurements.

Notebook: W01\_From\_a\_biological\_question\_to\_defensible\_evidence.ipynb

### Guided steps

1. Predict GC for ACGTGC before execution.
2. Run the calculation and then the edge-case tests.
3. Run your assigned synthetic sequence; change one A to G and predict the direction.
4. Save a short interpretation and your disclosure.

### Write your prediction

---

---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Use ACGTGC, GCNN and NN. Count with a pencil. Write 4/6, 2/2 and undefined, then complete the same three questions.

## W01 | Explain and submit

**Q1. State the numerator and denominator for ACGTGC and GCNN.**

---

---

---

**Q2. Report your sequence result and explain one test that would reveal a bug.**

---

---

---

**Q3. Can this GC result identify a species or establish that a sequence is biologically functional? Explain.**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: change the sequence by one base and predict how GC changes before running it.

Sources: S01, S02, S05, S10, S20, S23; full details in Course Source Register.

## W02 | Files, coordinates and reference versions

Validate interval lengths and identify missing reference metadata.

### Before you begin

A genome is a sequence; an annotation adds named features on a particular sequence version.

Notebook: W02\_Files\_\_coordinates\_and\_reference\_versions.ipynb

### Guided steps

1. Read each record and name its purpose.
2. Convert the BED interval to one-based inclusive coordinates.
3. Compare extracted bases and lengths.
4. Find the defect in the intentionally incorrect conversion.

### Write your prediction

---

---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Draw ten boxes labelled 1-10 above and 0-9 below. Shade positions 3-7. Check five bases GTACG.

## W02 | Explain and submit

**Q1. Explain why adding one to both BED endpoints changes the region.**

---

---

---

**Q2. Are BED [0,5) and [5,8) overlapping? Explain by enumerating bases.**

---

---

---

**Q3. What can you conclude about a reference build from a header containing only chr1?**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: inspect how a VCF deletion includes an anchor base; no indel normalization is examined.

Sources: S03, S04, S05; full details in Course Source Register.

## W03 | Reading and checking short Python programs

Diagnose a silent denominator error using small tests.

### Before you begin

GC fraction counts G and C bases among the bases your denominator actually includes.

Notebook: W03\_Reading\_and\_checking\_short\_Python\_programs.ipynb

### Guided steps

1. Trace the defective function by hand on GCNN.
2. Read the corrected function and explain each change.
3. Add one new input-output test.
4. State the biological meaning of the returned value.

### Write your prediction

---

---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Trace GCNN manually. The buggy fraction is  $\frac{2}{4}$ , the known-base fraction is  $\frac{2}{2}$ . Write three test cases.

## W03 | Explain and submit

**Q1. Why can a program run without an error and still be scientifically wrong?**

---

---

---

**Q2. Explain the difference between returning None and returning zero for NN.**

---

---

---

**Q3. Give your own test with input, expected output and the assumption being checked.**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: write a new test for lower-case letters and explain why it matters.

Sources: S05; full details in Course Source Register.

## W04 | Sequence similarity without overclaiming

Distinguish an alignment score from evidence of shared function.

### Before you begin

DNA sequences and protein sequences use different alphabets; compare like with like.

Notebook: W04\_Sequence\_similarity\_without\_overclaiming.ipynb

### Guided steps

1. Score a fixed alignment using +2 match, -1 mismatch.
2. Inspect a small exact local-alignment score matrix.
3. Change the mismatch penalty, not the biology.
4. Compare the score with a short evidence interpretation.

### Write your prediction

---

---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Write the two four-base sequences, mark three matches, calculate 5. Compare to a hypothetical 75%-identity match covering only four bases of a long protein.

## W04 | Explain and submit

**Q1. Show how the fixed alignment score of ACGT versus ACCT is calculated.**

---

---

---

**Q2. Why is 75% identity not a probability that two proteins have the same function?**

---

---

---

**Q3. What information would you report alongside a real BLAST hit before making a functional claim?**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: inspect affine gap penalties and substitution matrices; deriving log-odds is not examined.

Sources: S06, S26, S28; full details in Course Source Register.

## W05 | From sample to reads: designing an experiment

D1: compare affordable RNA-seq designs without confusing depth with replication.

### Before you begin

DNA sequencing measures sequence fragments; RNA sequencing samples molecules derived from RNA.

Notebook: W05\_From\_sample\_to\_reads\_\_designing\_an\_experiment.ipynb

D1 submission: completed notebook,  $\leq 1200$ -word design memo, one design table, GENOMES audit and AI/no-AI disclosure. Due end of Week 6; 12% of course. See project brief for rubric.

### Guided steps

1. Read the D1 brief before editing any parameter.
2. Compute cost and estimated usable fragments for three designs.
3. Compare independent cultures, balance, confounding and read depth.
4. Use the separate coverage example only to check units, not to set an RNA-seq threshold.

### Write your prediction

---



---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Calculate A:  $4 \times (150 + 800) = 3800$ ; B:  $8 \times (150 + 300) = 3600$ ; C:  $12 \times (150 + 150) = 3600$ . Usable pairs/sample: 64, 24, 12 million. Write a balanced batch-allocation table.

## W05 | Explain and submit

**Q1. Recommend A, B, C or a justified alternative and show the budget calculation.**

---

---

---

**Q2. Specify biological replicates, controls, allocation to batches and one exclusion rule fixed before inspecting outcomes.**

---

---

---

**Q3. Explain why the toy DNA mean-depth calculation neither guarantees per-base coverage nor proves sufficient RNA-seq power.**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: derive the ideal Poisson zero-coverage probability  $\exp(-C)$ , then explain why real coverage may violate that model.

Sources: S03, S10; full details in Course Source Register.

## W06 | Variation, independence and multiple testing

Use a known null simulation to distinguish false positives from posterior probabilities.

### Before you begin

Technical repeats share biological origins; more observations are not automatically more independent evidence.

Notebook: W06\_Variation\_independence\_and\_multiple\_testing.ipynb

### Guided steps

1. Adjust a four-test example by hand.
2. Run 500 all-null families of 100 uniform p-values.
3. Compare unadjusted counts with BH discoveries.
4. Explain the family-level interpretation rather than claiming every run contains exactly five errors.

### Write your prediction

---

---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Use the four p-values and multiply ordered values by  $4/\text{rank}$ , then impose reverse cumulative minima. For 100 tests, expected unadjusted errors = 5.

## W06 | Explain and submit

**Q1. Which of the four worked-example tests pass BH at 0.05? Show adjusted values.**

---

---

---

**Q2. Why is the expected number of unadjusted false positives  $100 \times 0.05$ , but not necessarily the observed number in a run?**

---

---

---

**Q3. Under this global null, why is FDR the probability of any discovery? Why is this not an individual gene probability?**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: show why dependence affects a multiple-testing guarantee; do not memorize a proof.

Sources: S07, S08, S10, S12; full details in Course Source Register.

## W07 | RNA counts, normalization and defensible expression claims

Keep sample identity intact and distinguish count normalization from inference.

### Before you begin

A gene may produce several transcripts; counts depend on what features and assignment rules were used.

Notebook: W07\_RNA\_counts\_\_normalization\_and\_defensible\_expression\_claims.ipynb

### Guided steps

1. Inspect the real two-gene excerpt and deliberately shuffled sample metadata.
2. Remove the documented fb suffix and explicitly match sample names.
3. Calculate a separate synthetic TPM example.
4. State which analyses the two-gene excerpt cannot support.

### Write your prediction

---



---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Match seven printed sample labels and conditions by name. Divide 100/1 and 200/2, then normalize each  $100/200 \times 1$  million.

## W07 | Explain and submit

**Q1. Explain why attaching condition labels by row order would be wrong here.**

---

---

---

**Q2. Why do counts 100 and 200 yield equal TPM in the synthetic example?**

---

---

---

**Q3. May the real two-gene excerpt support a valid DESeq2 analysis or be treated as the full library? Explain.**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: inspect median-of-ratios normalization and effective lengths; no dispersion derivation is assessed.

Sources: S09, S10, S25; full details in Course Source Register.

## W08 | Midterm review and the evidence audit

Audit six claims by distinguishing observations from unjustified conclusions.

### Before you begin

The midterm tests the floor: biology-to-data reasoning, coordinates, short code, depth and statistical interpretation.

Notebook: W08\_Midterm\_review\_and\_the\_evidence\_audit.ipynb

### Guided steps

1. Read the six fictional statements without using a chatbot.
2. Classify each as supported, overclaimed or incomplete.
3. Rewrite three statements with their evidence boundary.
4. Record one authoritative source for a correction.

### Write your prediction

---

---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Use the six printed claims. Mark each statement and cite one course source in writing; no internet is required if using the source register.

## W08 | Explain and submit

**Q1. Rewrite claims 1 and 2 accurately.**

---

---

---

**Q2. For claim 3, distinguish structural confidence from a measured binding endpoint.**

---

---

---

**Q3. Which claim is supported only within a clearly stated toy calculation, and what provenance would you still record?**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: exchange one evidence audit with a peer and check whether you agree on the scope.

Sources: S03, S04, S07, S10, S12, S23; full details in Course Source Register.

## W09 | Patterns, prediction and data leakage

D2: reveal donor leakage with a simple model and a defensible evaluation unit.

### Before you begin

A feature is a measured input; a label is the outcome being predicted. Rows from the same donor can share a signature.

Notebook: W09\_Patterns\_\_prediction\_and\_data\_leakage.ipynb

D2 submission: notebook,  $\leq 1200$ -word evaluation memo, split diagram/table, metrics and limitations, GENOMES audit and AI/no-AI disclosure. Due end Week 10; 12% of course. No advanced neural network required.

### Guided steps

1. Read the D2 scenario and predict which split is easier.
2. Compare within-donor row splitting with a new-donor holdout.
3. Report row and donor metrics, donor counts and the fitted scaling boundary.
4. Run donor-level label permutations as a reference distribution, not a clinical validation.

### Write your prediction

---



---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Use supplied sample outputs from the demo notebook solution as labelled simulation results: known-donor 1.00, new-donor value depends on seed. Explain the split diagram and count four independent holdout donors. Do not invent a real diagnostic study.

## W09 | Explain and submit

**Q1. What future population does each split actually evaluate, and where is leakage for the new-donor claim?**

---

---

---

**Q2. Report both metrics, donor counts and the null reference. Why is a four-donor test set too small for a stable clinical conclusion?**

---

---

---

**Q3. Propose a revised external evaluation with preprocessing, model selection and batch separation handled correctly.**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: nested cross-validation and calibration; no neural-network training is required.

Sources: S11, S13; full details in Course Source Register.

## W10 | Single cells and biological replication

Aggregate counts by donor and cell type without erasing biological replication.

### Before you begin

Bulk RNA-seq averages mixtures. Single-cell measurements can separate cell populations, but cells remain nested within samples.

Notebook: W10\_Single\_cells\_and\_biological\_replication.ipynb

### Guided steps

1. Inspect 32 synthetic cells from four donors and two cell types.
2. Sum counts within donor x cell-type groups.
3. Keep metadata for condition and donor.
4. Count independent donor replicates before and after aggregation.

### Write your prediction

---



---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Four cells with G1 counts 2,3,1,4 from one donor and cell type sum to 10. Repeat for another donor; keep two separate columns, not one condition sum.

## W10 | Explain and submit

**Q1. Explain why 32 cells do not become 32 independent donors.**

---

---

---

**Q2. Which grouping keys preserve donor-level replication for a within-cell-type condition comparison?**

---

---

---

**Q3. Does successful aggregation establish adequate power or remove all batch confounding? Explain.**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: differential abundance and paired donor designs; no atlas integration pipeline is required.

Sources: S09, S12, S13, S27; full details in Course Source Register.

## W11 | Biological AI and the honest model scorecard

Build a model evidence scorecard without running an expensive model.

### Before you begin

Models see numerical representations of biological measurements. A useful representation does not itself establish a mechanism.

Notebook: `W11_Biological_AI_and_the_honest_model_scorecard.ipynb`

### Guided steps

1. Read the two real-study evidence cards.
2. Separate zero-shot use from fine-tuning.
3. Identify task, baseline, split and evidence limits.
4. Draft a small fair evaluation plan with one simple baseline.

### Write your prediction

---

---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Complete a seven-field scorecard on paper: task,data,split,adaptation,baseline,metric,limitation. Use the two source summaries.

## W11 | Explain and submit

**Q1. Why are apparently different conclusions in the two cards not automatically contradictory?**

---

---

---

**Q2. What baseline, holdout unit and metric would you require for your stated task?**

---

---

---

**Q3. Name one result that is unknown from these cards and must not be invented.**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: sketch a domain-shift stress test without implementing a foundation model.

Sources: S11, S14, S15; full details in Course Source Register.

## W12 | Protein structures: prediction, confidence and function

Interpret local and relative structural confidence without inferring binding.

### Before you begin

A protein is an amino-acid chain; its structure and interactions depend on sequence and biological context.

Notebook: W12\_Protein\_structures\_\_prediction\_\_confidence\_and\_function.ipynb

### Guided steps

1. Read the real hemoglobin model evidence card.
2. Inspect a separate synthetic two-domain confidence example.
3. Locate low-confidence residues and uncertain relative placement.
4. Choose a validation experiment for a stated functional claim.

### Write your prediction

---



---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Circle residues 6-8 in the synthetic list. Compare PAE diagonal 2,3A with off-diagonal 18,20A. Write confidence versus function as two separate statements.

## W12 | Explain and submit

**Q1. What does the high pLDDT in the real card support, and what does it not establish?**

---

---

---

**Q2. Interpret high off-diagonal PAE with low within-domain PAE in the synthetic example.**

---

---

---

**Q3. Does low pLDDT prove intrinsic disorder? What would you do before making a functional claim?**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: inspect a predicted complex with interface metrics; no docking or molecular dynamics is required.

Sources: S01, S16, S17, S24; full details in Course Source Register.

## W13 | From a variant to an evidence-based sentence

Write a bounded variant interpretation using fictional evidence.

### Before you begin

A sequence difference is a variant. Its effect depends on location, transcript, mechanism and biological context.

Notebook: W13\_From\_a\_variant\_to\_an\_evidence\_based\_sentence.ipynb

### Guided steps

1. Read the three fictional variants without identifying a patient.
2. Distinguish review status, clinical context and evidence strength.
3. Recognize a VUS and conflicting evidence.
4. Write a cautious report sentence and identify what evidence is missing.

### Write your prediction

---



---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Use the three fictional cards. Underline uncertain/conflicting/expert-reviewed, then separate review quality from certainty in a report sentence.

## W13 | Explain and submit

**Q1. Write a two-sentence report for TEACH-A without recommending treatment.**

---

---

---

**Q2. Why does expert-panel review not mean a 99% or 100% pathogenicity probability for every clinical context?**

---

---

---

**Q3. What identifiers and contextual evidence would you seek before evaluating an actual variant?**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: explore current gene-specific ClinGen specifications as a supervised follow-on topic.

Sources: S03, S18, S19; full details in Course Source Register.

## W14 | Responsible bioinformatics and your scientific record

Complete a responsible release decision and a reproducible course portfolio.

### Before you begin

Genetic and health information can be sensitive even when a name column has been removed.

Notebook: W14\_Responsible\_bioinformatics\_and\_your\_scientific\_record.ipynb

### Guided steps

1. Classify five fictional sharing scenarios.
2. Identify data sensitivity, purpose and permitted destination.
3. Complete the GENOMES record for one project.
4. Prepare a release decision and an honest limitation statement.

### Write your prediction

---



---

### Record the result

| Item                         | Your observation |
|------------------------------|------------------|
| Data status and source       |                  |
| Key number / table and units |                  |
| Independent biological unit  |                  |
| Check performed and outcome  |                  |

### Device-free alternative

Write a three-column table: scenario,allow/pause,reason. Complete the GENOMES card on your printed D1 or D2 report.

## W14 | Explain and submit

**Q1. Which scenarios need a pause and an authorized institutional review, and why?**

---

---

---

**Q2. Explain the difference between pseudonymization and a contextual claim of anonymization.**

---

---

---

**Q3. Write your project release statement: what may be shared, what remains uncertain, who is responsible, and what AI use occurred?**

---

---

---

### Scope and disclosure

One thing these results cannot establish:

---

AI / help disclosure, including "No AI used" when applicable:

---

Before submission: include your course pseudonym, result or documented failure, substantive answers and an actual check. A completed field is not proof that the answer is correct. Use the project brief rather than this worksheet alone for D1 and D2.

Optional extension: Optional: compare controlled-access and federated analysis designs without moving real patient data.

Sources: S20, S21, S22, S23; full details in Course Source Register.